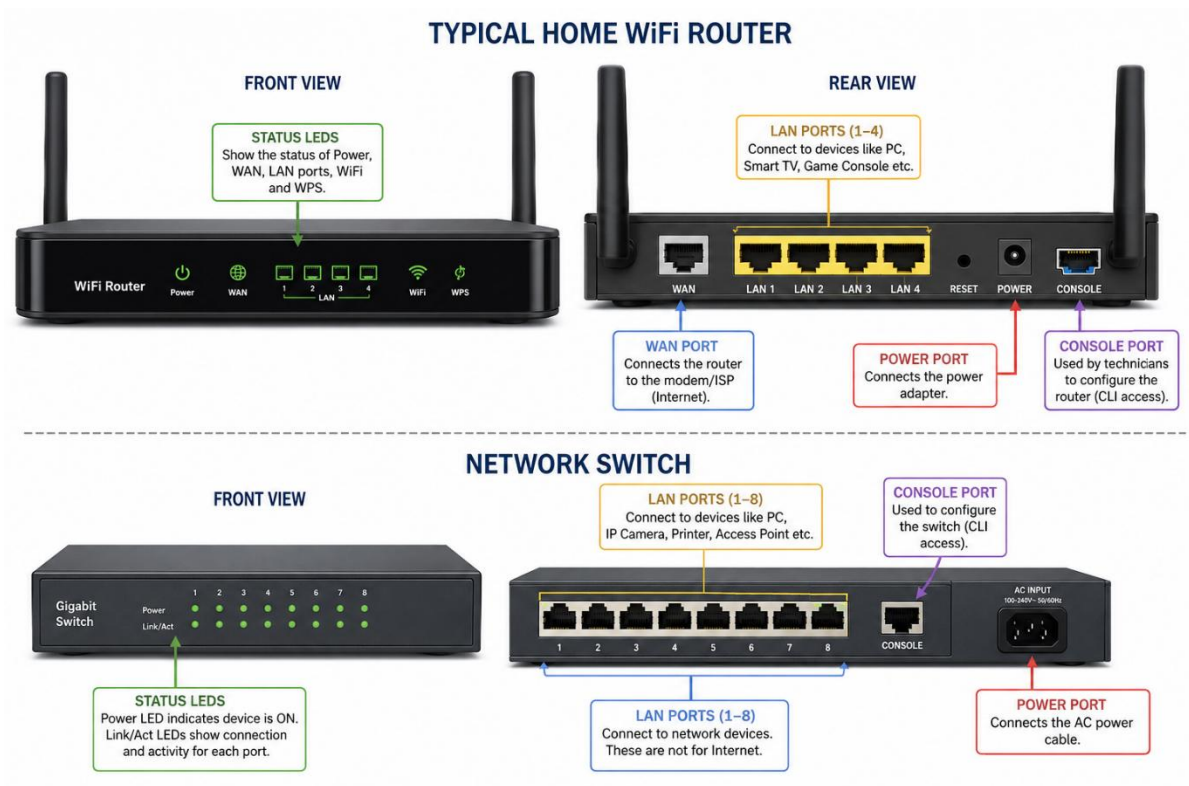


1) Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs



2) List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic, think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect

| Difference          | Router                                                                               | Switch                                                                             |
|---------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1. Main Function    | Connects <b>different networks</b> together, such as a home network to the Internet. | Connects <b>devices within the same local network (LAN)</b> .                      |
| 2. Addresses Used   | Mainly uses <b>IP addresses</b> to determine where to send packets                   | Mainly uses <b>MAC addresses</b> to forward Ethernet frames to the correct device. |
| 3. Traffic Handling |                                                                                      |                                                                                    |

3) Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

#### Router

| Component          | Position                                   | Purpose                                                                   |
|--------------------|--------------------------------------------|---------------------------------------------------------------------------|
| Power Port         | Rear side, usually near the right side     | Connects the power adapter to supply electricity to the router.           |
| Ethernet/WAN Port  | Rear side, usually separate from LAN ports | Connects the router to the modem/Internet connection.                     |
| Ethernet/LAN Ports | Rear side, commonly 4 ports                | Connect computers, TVs, printers, or other wired devices.                 |
| LED Indicators     | Front panel                                | Show Power, Internet/WAN, LAN activity, Wi-Fi, and other operating status |

#### Network Switch

| Component      | Position                                             | Purpose                                                      |
|----------------|------------------------------------------------------|--------------------------------------------------------------|
| Power Port     | Rear side                                            | Connects the switch to an electrical power source.           |
| Ethernet Ports | Rear side, usually separate from LAN ports           | Front or rear panel, commonly 5, 8, 16, 24, etc.             |
| LED Indicators | Front panel, usually above/beside each Ethernet port | Show power, link status, and network activity for each port. |

**4) Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.**

For a small gaming café, I would connect the ISP line to the router first. The router connects the café's local network to the Internet and manages IP addresses and Internet traffic. Next, I would connect one of the router's LAN ports to a network switch using an Ethernet cable. Then, I would connect all 8 gaming PCs to the switch using separate Ethernet cables.

Connection:

ISP → Router → Network Switch → 8 Gaming PCs

I would use a switch because it provides enough Ethernet ports for all 8 PCs and allows them to communicate efficiently on the local network. A wired Ethernet connection is preferable for multiplayer gaming and IPL streaming because it generally provides lower latency and more stable connectivity than Wi-Fi.